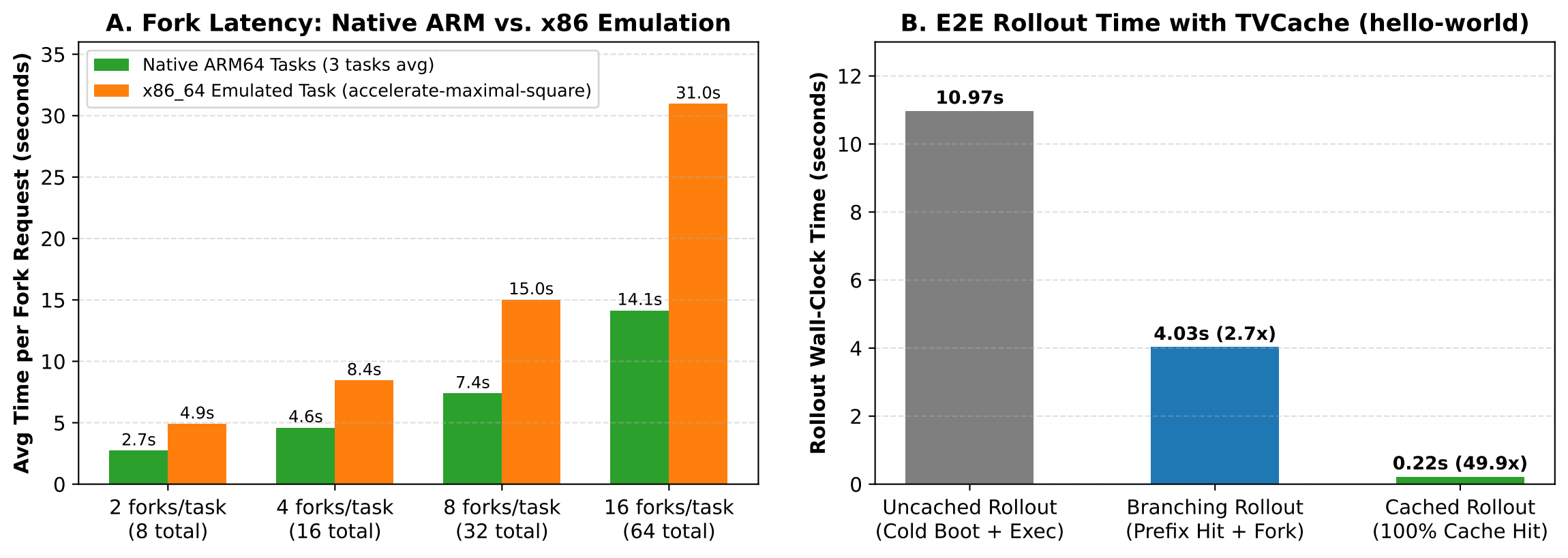
A. Fork Latency: Native ARM vs. x86 Emulation
Native ARM64 Tasks (3 tasks avg)
x86_64 Emulated Task (accelerate-maximal-square)
Avg Time per Fork Request (seconds)
0
5
10
15
20
25
30
35
2.7s
4.9s
4.6s
8.4s
7.4s
15.0s
14.1s
31.0s
2 forks/task
(8 total)
4 forks/task
(16 total)
8 forks/task
(32 total)
16 forks/task
(64 total)
B. E2E Rollout Time with TVCache (hello-world)
Rollout Wall-Clock Time (seconds)
0
2
4
6
8
10
12
10.97s
4.03s (2.7x)
0.22s (49.9x)
Uncached Rollout
(Cold Boot + Exec)
Branching Rollout
(Prefix Hit + Fork)
Cached Rollout
(100% Cache Hit)